WNDRVR
SUPPORT NETWORK

VXWORKS 7 OUTSIDE LAYERS TUTORIAL, SR0650

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VxWorks 7 Outside Layers Tutorial, SR0650

10 June 2020

1. OUTSIDE LAYERS TUTORIAL

This VxWorks ® tutorial provides instructions on how to create, use and share custom layers that reside outside the Wind River installation directory.

VxWorks 7 introduces the layer mechanism as a clean way to encapsulate functionality like software applications, BSPs and Drivers. Layers are added and removed from a VxWorks system without any risk of damage to that system. The layers can reside inside or outside the Wind River installation. This tutorial explores the use of layers outside the Wind River installation.

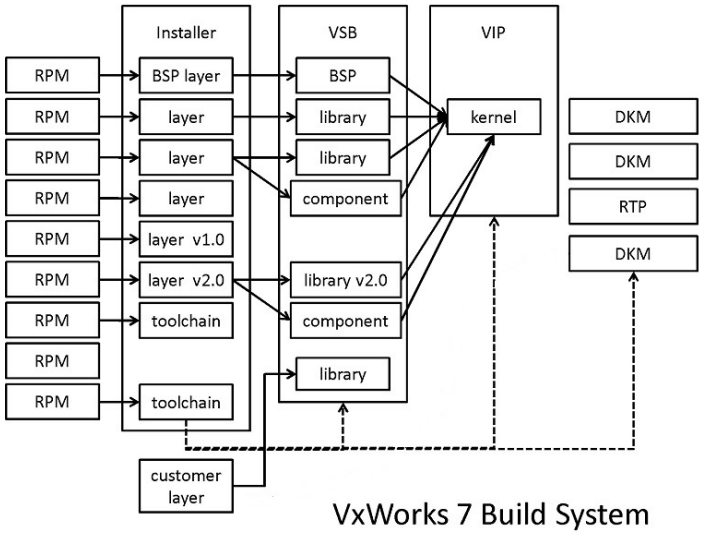
For this tutorial the workflow is as follows:

1. Create a layer.
2. Add the layer to a VSB and testing it with a VIP.
3. Export the layer in an RPM.
4. Import the layer RPM.
5. Uninstall the layer.

2. INSIDE AND OUTSIDE LAYERS

VxWorks software components are delivered as RPM files, packaged as layers and may reside inside or outside of the Wind River installation directory.

Source and binary components are presented to the VxWorks build system as RPM files containing layers. A layer encapsulates a piece of functionality that may be built into the VxWorks kernel.



Prior to VxWorks product installation, the functionality within each layer is packaged as an RPM supplied either on installation media, or retrieved from the Wind River servers by the installer or **Product Maintenance** tool. At fresh installation time, the installer takes the RPM files, unpacks them and assembles the initial set of VxWorks layers. The **Product Maintenance** uses the installer to incrementally add new layers into the Wind River installation after a fresh product installation has completed. This is a useful mechanism for both partners and customers who wish to easily add new functionality to VxWorks like a BSP, an API, or a driver.

Layers that exist within the installation directory are located at *installDir* **/vxworks-7/pkgs_v2/**. These layers are referred to as *inside layers*, because they are positioned inside the installation directory structure.

Layers are also a useful way of encapsulating software applications which have been developed by a Wind River customer. However, this customer may not want to be forced to deliver their software application to their own end customers as an internal layer. They may have good reason to decouple their own product installation from the Wind River installation. For this reason, layers may also be positioned outside of the installation. Layers can exist somewhere else on the file system, external to *installDir* **/vxworks-7/pkgs_v2/**. These layers are referred to as *outside layers* because they are located outside of the installation directory.

Outside Layers have all the features and benefits of inside layers except one. The installer cannot install the outside layer to the file system, it can only tell the VxWorks build system that the outside layer is expected to reside there. It is therefore your responsibility to ensure that the RPM file is extracted to the correct directory before the installer reads the RPM and identifies it as an outside layer.

For example tutorials that lead you through the creation of inside and outside layers, please refer to VxWorks 7 Inside Layers Tutorial and VxWorks 7 Outside Layers Tutorial and VxWorks 7 Layer Version Management Tutorial

3. EXAMPLE: CREATING AN OUTSIDE CUSTOM LAYER AND INTEGRATING IT INTO A VSB

You can create a custom layer externally to the VxWorks source tree and integrate it into your VxWorks source build (VSB) project using the project build system.

Before You Begin

This example assumes the Workbench project explorer already contains a **cobble** downloadable kernel module (DKM) example project.

It also assumes you have a pre-existing VxWorks source build (VSB) and VxWorks image project (VIP) present in Workbench. Follow the steps below using a VxWorks simulator (VxSim) instance as the target.

This tutorial also assumes you have no previously created **cobbleLayer** projects, such as the one developed in **Example: Creating an Inside Custom Layer and Integrating it into a VSB**. Before you begin, remove any existing **cobbleLayer** projects, if necessary, by deleting them from the filesystem.

About This Task

For this tutorial, you use the Workbench **cobble** example DKM project to represent the application code that you want to integrate into the layer. Create a layer project, insert the **cobble** DKM application code into it, rebuild the VSB and VIP, and check that the target boots and runs.

Procedure

1. Create a folder for your new layer project.

   In this example, you are creating an outside custom layer. Therefore, the path to this folder must be outside the Wind River installation. The layer project folder should not be positioned within *installDir* **/VxWorks-7/pkgs_v2**. Instead, it belongs externally; for example, in **C:/MyLayers/cobbleLayer**.

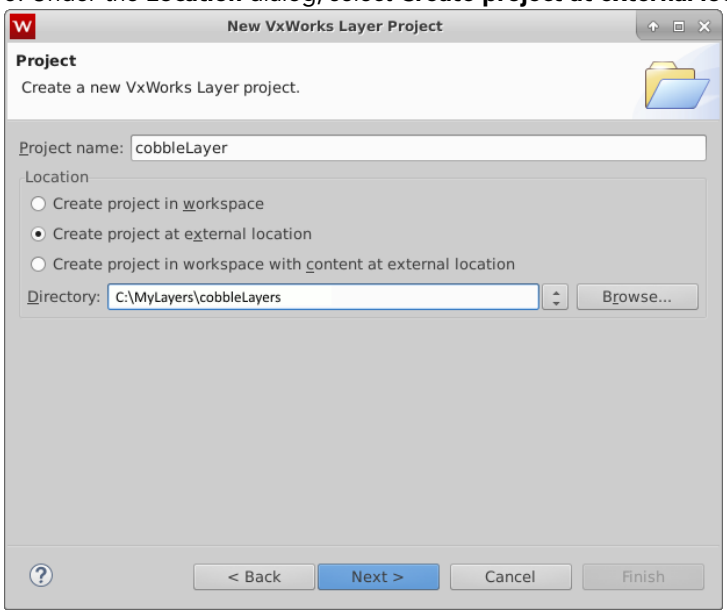
2. Edit the *WIND_LAYER_PATHS* environment variable and add the full path to your new layer project folder. This layer path is **C:\MyLayers\cobbleLayer**.
   1. Go to **Control Panel > System**.
   2. In the left pane, click on **Advanced system settings**.
   3. On the Advanced pane, click **Environment Variables**.
   4. In the **System Variables** area, add a new variable called *WIND_LAYER_PATHS*.
   5. Set this new variable to your custom path. Click OK. Save and exit as appropriate.

   Note: If *WIND_LAYER_PATHS* exists, and already contains a path, append a semicolon to the pre-existing path and insert the new custom path after the semicolon for Windows systems. For Linux systems, use colon as path separators. Do not leave space before the new custom path.

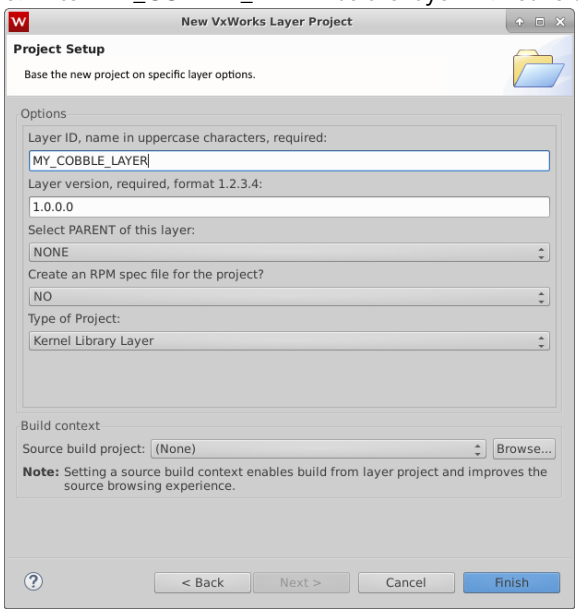
   Note: You may need to reboot your host workstation in order for the *WIND_LAYER_PATHS* environment variable change to take place.

3. Create a VxWorks layer project in your new layer project folder and call it **cobbleLayer**.
   1. From the Workbench menu, select **File > New > Wind River Workbench Project**.
   2. Select the option VxWorks 7 as the target operating system from the drop-down menu if you are prompted to choose the **Target Operating System**.

3. From the Build type drop down list, select **Layer**. Click **Next**.
4. Name the Project **cobbleLayer**.
5. Under the **Location** dialog, select **Create project at external location** and browse to the layer folder. Click **Next**.



6. Enter **MY_COBBLE_LAYER** as the layer ID. Leave all other options as default and click **Finish**.



4. Populate **cobbleLayer** with **cobble.c**.
    1. In the Workbench project explorer, expand the **cobble** project.
    2. Right-click on **cobble.c** and select **Copy**.
    3. In the project explorer, navigate to **cobbleLayer**. Highlight the **src** folder within the **cobbleLayer** project and click **Paste**.
5. Use a host terminal to check that **cobbleLayer** exists.

```
C:\WindRiver\workspace> wrenv -p vxworks-7
C:\WindRiver\> wrtool -data C:\WindRiver\workspace
Active workspace is 'C:\WindRiver\workspace'.
WrTool, version 4.1.0
```

```
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workspace> prj vsb info MY_COBBLE_LAYER
Name:                       MY_COBBLE_LAYER
VERSION:                    1.0.0.0
[...]
Location: C:/MyLayers/cobbleLayer
```
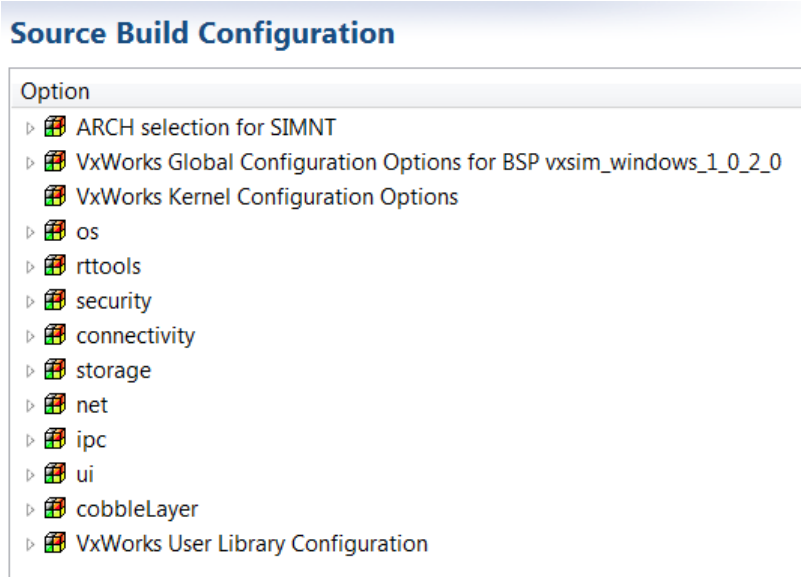

6. Verify that your VSB has access to **cobbleLayer** by opening the VSB source build configuration in the project explorer.

   You can see **cobbleLayer** appearing there as a root component in the Source Build Configuration.



7. Include **cobbleLayer** in your VSB.
   1. Right-click on the red component within **cobbleLayer** and select **Add with Dependencies** to add it to the VSB.
   2. In the project explorer, right-click on the VSB and select **Rebuild Project**.
8. Modify your VIP to start the cobble application upon target startup.
   1. Locate the VIP related to your VSB in the project explorer.
   2. Locate the **usrAppInit.c** file within the project and open it for editing.
   3. Insert a call to the **cobble.c** progStart( ) function into the **usrAppInit.c** usrAppInit( ) function.

```
/*******************************************************************************
*
* usrAppInit - initialize the users application
*/

void usrAppInit (void)
    {
#ifdef  USER_APPL_INIT
    USER_APPL_INIT;     /* for backwards compatibility */
#endif

    /* TODO: add application specific code here */
    progStart();
    }
```

9. Rebuild the VIP and reboot the target from the rebuilt VIP. Check that the expected cobble tasks are running on the target from the VxWorks shell.

   The **cobble.c** file is a simple multi-tasking application that simulates a data collection system in which data is retrieved from an external source (for example, a device that generates interrupts as data comes in). The four cobble tasks are:

- tCosmos, which simulates an interrupt service routine (ISR) and generates new data.
- tSchlep, which collects the data.
- tCrunch, which performs calculations on the data to obtain a result value.
- tMonitor, which monitors the results and prints a warning message if they are outside of the safety range.

> **Note:** If you do not add progStart( ) to **usrAppInit.c**, cobble does not start upon target reboot and **cobbleLayer** does not link into the kernel image. Rather, the layer is omitted by the VxWorks build system because there is no reference to the **cobble.c** application in the VxWorks BSP or application code. As a consequence, the VxWorks build system linker omits **cobbleLayer** during the link phase of the VIP build.

```
-> i

  NAME         ENTRY       TID    PRI   STATUS      PC       SP     ERRNO  DELAY
----------  ------------ -------- --- ---------- -------- -------- ------- -----
tIsr0       102678f0     3800da08   0 PEND       102714ee 380cff60       0     0
tJobTask    101a82b0     38049e08   0 PEND       102714ee 3813ff00       0     0
tExcTask    101a7ca0     103dfb40   0 PEND       102714ee 103e5d40       0     0
tLogTask    logTask      3804dd48   0 PEND       1026f073 3817feb0       0     0
tShell0     shellTask    38369d40   1 READY      1027a760 3841fc40       0     0
tErfTask    100ed8c0     38002e10  10 PEND       10271e1f 3802ff00       0     0
ipcom_tick> 1003a1b0     38437e48  20 PEND       102714ee 384dfed0       0     0
tVxdbgTask  100ffc70     3832b4b0  25 PEND       102714ee 3838ff20       0     0
tTcfEvents  1020f1f0     103f8380  49 PEND+T     102714ee 103f81f0       0    54
tTcf        1015a0b0     383ffb18  49 PEND       102714ee 3845f890       0     0
tTcf        1015a0b0     38433a10  49 PEND       102714ee 3849f8a0       0     0
tTcf        1015a0b0     38479908  49 PEND       102714ee 3861f8f0       0     0
tTcf        1015a0b0     3847d5c8  49 PEND       102714ee 3865fa40       0     0
tNet0       ipcomNetTask 38008b30  50 PEND       102714ee 3806fef0       0     0
tAioIoTask> aioIoTask    3811b440  50 PEND       10271e1f 381ffef0       0     0
tAioIoTask> aioIoTask    3811e870  50 PEND       10271e1f 3823fef0       0     0
ipcom_sysl> 10038c10     38086280  50 PEND       10271e1f 382bfe20       0     0
tNetConf    10075840     380a5ec8  50 PEND       102714ee 3834fcf0       0     0
tAnalysisA> cafe_event_> 103d9ca0  50 PEND       102714ee 103d9b30       0     0
tAioWait    aioWaitTask  38117f10  51 PEND       102714ee 381bfe80       0     0
tCosmos     cosmos       38438518 200 DELAY      10279812 3851ff80       0     1
tSchlep     schlep       38438948 220 PEND       102714ee 3855ff20       0     0
tMonitor    monitor      3843f900 230 READY      10149800 385dffb0       0     0
tCrunch     crunch       3843f5d0 240 READY      100b5ae0 3859ffd0       0     0
ipcom_egd   10037c10     38089998 255 READY      10037c80 382ffec0       0     0
value = 0 = 0x0
->
```

4. EXAMPLE: EXPORTING AN OUTSIDE CUSTOM LAYER AS AN RPM

VxWorks Layer Projects are portable, and can be shared in the form of an RPM.

Before You Begin

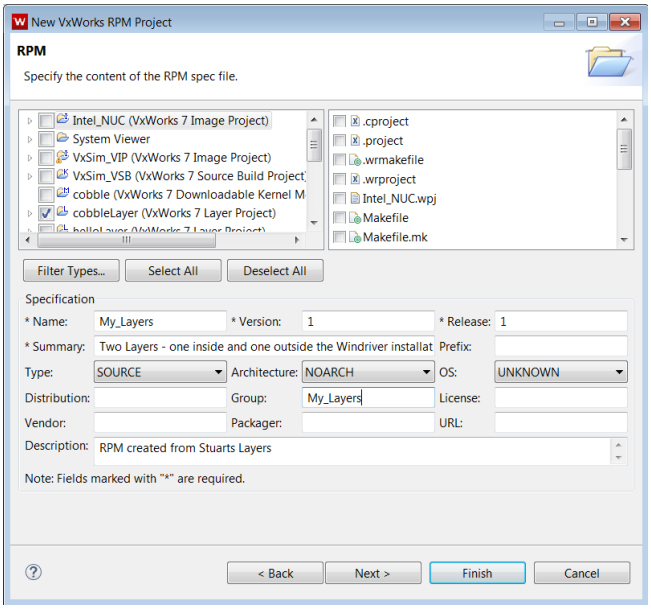
This example assumes you have already completed the Example: Creating an Outside Custom Layer and Integrating it into a VSB on page 3. That example results in a VxWorks layer project called **cobbleLayer** that is located outside the Wind River installation directory and also present in the Workbench project explorer.

About This Task

Now create an RPM file in this example which contains **cobbleLayer**. The generated RPM file is sharable with other VxWorks developers who need to build **cobbleLayer** into their VSB.

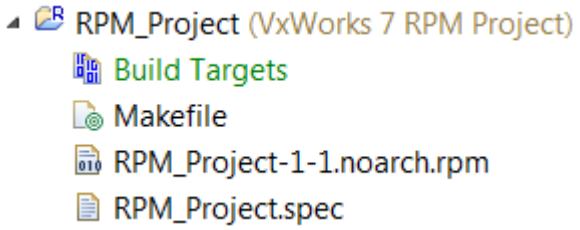
Procedure

1. Create an RPM Project and name it.
    1. From the Workbench menu, select **File > New Wind River Workbench Project**.
    2. Select the option VxWorks 7 as the target operating system from the drop-down menu if you are prompted to choose the **Target Operating System**.
    3. As Build Type, select **RPM Project**.
    4. Name the project **RPM_Project** then click **Next**.
2. Give **RPM_Project** a payload.
    1. Select the **cobbleLayer** project as the RPM payload.
    2. Name the project **My_Layers** and also specify the Group **My_Layers**. Leave the other fields as default.



3. Click **Finish**.
4. Generate the RPM file by selecting right-click on **RPM_Project** in the Workbench project explorer and selecting **Build Project**.

The project build will result in two files. The file **RPM_Project-1-1.noarch.rpm** is the portable RPM file that may be imported into a fresh Wind River installation on an additional Workstation.



5. Prepare to share **cobbleLayer**.
   1. Create a copy of the directory **C:\MyLayers\cobbleLayer**.
   2. Export **cobbleLayer** by transferring **C:\MyLayers\cobbleLayer** and **RPM_Project-1-1.noarch.rpm** to a destination workstation. For further details, refer to Example: Importing an Outside Custom Layer as an RPM on page 9.

Note: It might seem redundant to copy the **cobbleLayer** project directory and share it along with **RPM_Project-1-1.noarch.rpm**? The function of the **RPM_Project-1-1.noarch.rpm** file is to encapsulate layer project directory and files.

While the RPM file does contain the **cobbleLayer** directory and files, the Wind River installer is only capable of installing VxWorks layer project files and directories into the Wind River directory structure. It cannot be used to install software outside of the root Wind River installation directory, although it can be used to inform the Workbench build system that those outside layers exist. Consequently, the outside layer is exported by manually copying **C:\MyLayer\cobbleLayer** to its destination before **RPM_Project-1-1.noarch.rpm** is imported into Workbench.

5. EXAMPLE: IMPORTING AN OUTSIDE CUSTOM LAYER AS AN RPM

VxWorks layer projects are portable, and when they are shared as RPM files, they are easily installed into fresh Wind River Workbench development environments.

Before You Begin

This example assumes you have already completed Example: Exporting an Outside Custom Layer as an RPM on page 7. You must not attempt this example on the workstation that was used with the previous example. This example must be followed on a new, fresh destination workstation containing a fresh installation of Wind River VxWorks 7.

The Workbench project explorer on the destination workstation should contain a VxWorks source build (VSB) project and a VxWorks image project (VIP).

The RPM file to be imported to the destination workstation must be copied to *installDir* **\workspace\RPM_Project-1-1.noarch.rpm** on the destination file system. Also, copy the **cobbleLayer** project directory from the original source workstation to the destination workstation.

You must also extract the VxWorks 7 installer into the destination workstation's root Wind River directory. If the installer has already been unzipped into the root Wind River folder, there will be a folder named *installDir* **\installer** and an executeable program named *installDir* **\setupc.exe**.

> Note: The VxWorks 7 installer is not extracted into the destination workstation's root Wind River directory at installation time. Do this manually once installation of the VxWorks 7 product has completed.
>
> The Wind River installer is located on the Wind River server or the source media you used to originally install the product.

About This Task

With this example you will install **cobbleLayer**, previously developed on a source workstation, into a fresh destination workstation. You will import and build **cobbleLayer** into the destination workstation's VSB and modify the related VIP so that **cobbleLayer** starts automatically on target reboot.

You will perform all steps below on the destination workstation, not on the original source workstation.

Procedure

1. Manually copy the **cobbleLayer** project folder to **C:/MyLayers/cobbleLayer** on the destination workstation.

   > Note: It may seem redundant to manually copy the **cobbleLayer** project directory when we are also importing **RPM_Project-1-1.noarch.rpm** that already contains the **cobbleLayer** directory and files. You do this because the Wind River installer is only able to install VxWorks layers into the Wind River installation directory structure. It cannot be used to install files externally on the workstation file system.

2. Install **RPM_Project-1-1.noarch.rpm** into Workbench from a terminal window.

```
cd c:\WindRiver
setupc.exe -installpath c:\WindRiver -installerUpdateURLs none -yum -y localinstall c:\Wind Riv
er\workspace\RPM_Project-1-1.noarch.rpm
```

> **Note:** This step notifies Workbench that the outside layer is present on the file system.
>
> Envoke setupc.exe and not setup.exe. Only the setupc.exe command window will remain open once the installation has completed.

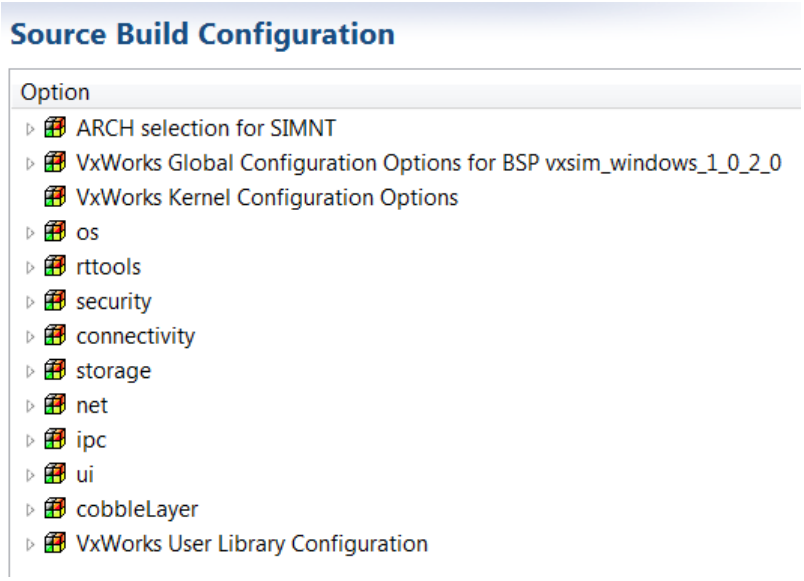
3. Edit the *WIND_LAYER_PATHS* environment variable and add the full path to **cobbleLayer**. The installation path should match the directory that was created in Step 1, e.g. **C:\MyLayers\cobbleLayer**.
   1. Go to **Control Panel** --> **System**.
   2. In the left pane, click on **Advanced system settings**.
   3. On the Advanced pane, click **Environment Variables**.
   4. In the **System Variables** area, add a new variable called *WIND_LAYER_PATHS*.
   5. Set this new variable to your custom path. Click OK and save and exit as appropriate.
4. Use a host terminal to verify correct **cobbleLayer** installation.

```
cd c:\WindRiver
C:\WindRiver\> setupc.exe -yum list installed
Default install directory: C:\WindRiver
Enter install directory or press enter
>
Launching installer...
Searching updates.windriver.com for installer updates.
C:\WindRiver\> ####################
C:\WindRiver\> Installed Packages
C:\WindRiver\> RPM_Layer.noarch  1-1  installed
```

> **Note:**
>
> Envoke setupc.exe and not setup.exe. Only the setupc.exe command window will remain open once the command has completed.

5. Verify that **cobbleLayer** is available to your VSB by opening the VSB source build configuration in the project explorer.

   You can see **cobbleLayer** appearing there as a root component in the Source Build Configuration.



6. Test the **cobbleLayer** you have imported by continuing from Step 7 of the Example: Creating an Outside Custom Layer and Integrating it into a VSB on page 3.

6. EXAMPLE: UNINSTALLING AN OUTSIDE CUSTOM LAYER RPM

VxWorks custom layer projects, when installed as RPM files, can be uninstalled without adversely affecting the Wind River installation.

Before You Begin

This example assumes that you have already completed the Example: Importing an Outside Custom Layer as an RPM on page 9.

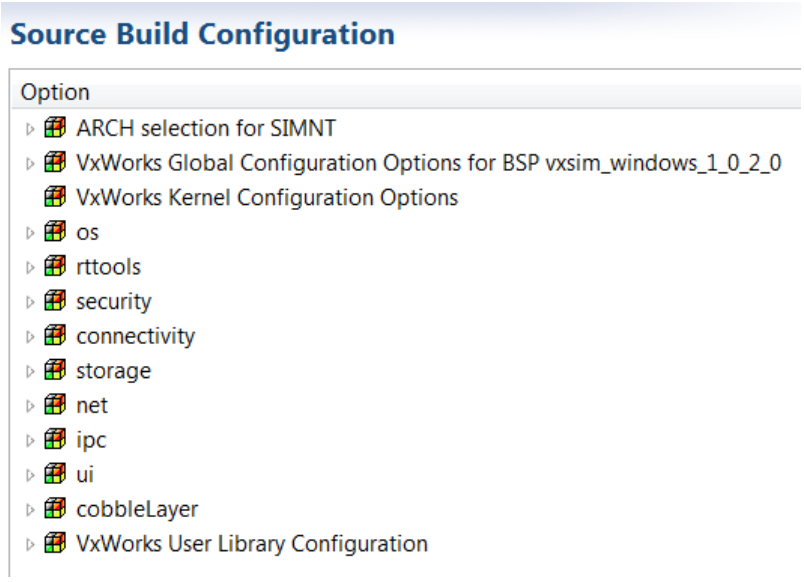
About This Task

A custom layer, **cobbleLayer**, has been installed into Wind River Workbench. You will uninstall **cobbleLayer** in this example.

Procedure

1. Verify that **cobbleLayer** is installed as a custom layer by opening the VSB Source Build Configuration in the project explorer.

   You can see **cobbleLayer** appearing there as a root component in the Source Build Configuration.



2. Uninstall **RPM_Project-1-1.noarch.rpm** from a host shell.

```
cd c:\WindRiver
```

```
setupc.exe -yum remove RPM_Project
```

> Note: Envoke setupc.exe and not setup.exe. Only the setupc.exe command window will remain open once the installation has completed.

3. Delete the layer directory at **C:\MyLayers\cobbleLayer**.

> **Note:** Because the **cobbleLayer** project directory is positioned externally to the Wind River installation directory, then both this directory and the *WIND_LAYER_PATHS* environment variable must be cleaned up manually.

4. Remove the reference to **cobbleLayer** from *WIND_LAYER_PATHS* environment variable.
   1. Go to **Control Panel** --> **System**.
   2. In the left pane, click on **Advanced system settings**.
   3. On the Advanced pane, click **Environment Variables**.
   4. In the **System Variables** area, locate the variable called *WIND_LAYER_PATHS*.
   5. Remove **C:\MyLayers\cobbleLayer** from the path. If *WIND_LAYER_PATHS* contains other paths once **C:\MyLayers\cobbleLayer** has been deleted, ensure that a semicolon exists between all the remaining directory paths. Do not put a space after a semicolon. Click OK and save and exit as appropriate.
5. Verify that **cobbleLayer** is no longer a layer component in Workbench.
   1. In the Workbench project explorer, open the VSB Source Build Configuration.
   2. Navigate to the bottom of the Source Build Configuration. Notice that **cobbleLayer** no longer appears as a root component.
6. **cobbleLayer** is completely uninstalled.

7. EXAMPLE: CREATING A OUTSIDE CUSTOM SUBLAYER TO REPLACE A KERNEL SUBLAYER

You can create a custom outside sublayer to replace a kernel sublayer if you want to keep the changes to a sublayer outside of the VxWorks 7 installation and under source control.

About This Task

For this example, a kernel sub-layer named **nfs** needs to be modified outside of the installation.

Procedure

1. Edit the *WIND_LAYER_PATHS* environment variable to add **myDir**. For more information, see step 2 of Example: Creating an Outside Custom Layer and Integrating it into a VSB on page 3.
2. Copy the layer to the directory you want to keep under version control.

   ```
   cp -r vxworks-7/pkgs_v2/storage/fs-2.1.0.1/nfs-2.0.1.1 myDir\mynfs
   ```

3. Edit the **layer.vsbl** file in **myDir\mynfs** by adding a fifth digit to the version number (for example, **VERSION 2.0.1.1.1**)
4. Find the parent of the sublayer **nfs-2.0.1.1** by typing the following command in a VxWorks Development Shell.

   ```
   vxprj vsb listAll |grep NFS
   ```

   This makes **FS_2_1_0_1_NFS_2_0_1_1** the full name of **nfs-2.0.1.1**. The **PARENT** is the first part of the name: **FS_2_1_0_1**.

5. Edit the **layer.vsbl** file in **myDir\mynfs** and add a reference to the parent directory.

   PARENT FS_2_1_0_1

6. You can now modify **\mynfs** and the VxWorks 7 installation remains unchanged.

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